

Activation Quantization

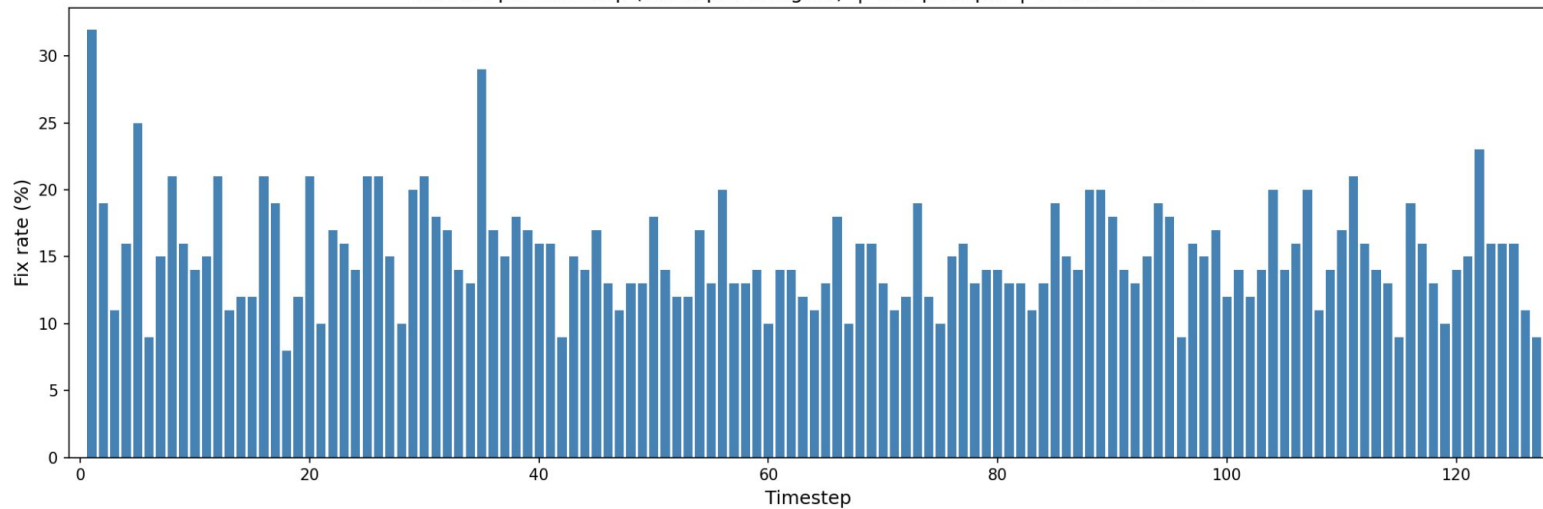
Ahmad Fraij

Issues with SINQ Quantized Model

```
LLaDAModelLM(  
  (model): LLaDAModel(  
    (transformer): ModuleDict(  
      (wte): Embedding(126464, 4096)  
      (emb_drop): Dropout(p=0.0, inplace=False)  
      (ln_f): RMSLayerNorm()  
      (blocks): ModuleList(  
        (0-31): 32 x LLaDALlamaBlock(  
          (dropout): Dropout(p=0.0, inplace=False)  
          (act): SiLU()  
          (attn_out): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)  
          (ff_out): SINQLinear(in_features=12288, out_features=4096, bias=False, device=cuda, W_q.device=cpu)  
          (rotary_emb): RotaryEmbedding()  
          (attn_norm): RMSLayerNorm()  
          (ff_norm): RMSLayerNorm()  
          (q_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)  
          (k_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)  
          (v_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)  
          (ff_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cuda, W_q.device=cpu)  
          (up_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cuda, W_q.device=cpu)  
        )  
      )  
      (ff_out): SINQLinear(in_features=4096, out_features=126464, bias=False, device=cuda, W_q.device=cpu)  
    )  
  )  
)
```

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        )  
      )  
      (ff_out): Linear(in_features=4096, out_features=126464, bias=False)  
    )  
  )  
)
```

Fix rate per timestep (val or pos disagree) | 100 prompts | overall = 15.0%



AWQ Quantization and Evaluation

- Quantized Llada and Dream models using AWQ and using 128 samples from the pile dataset with max sequence of 2048

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The first/second number indicates the % of correct/incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ)	1.88 / 1.28	1.60 / 1.30	1.03 / 0.49	2.84 / 2.84	3.85 / 4.37	0.64 / 0.28
LLaDA-8B-Base (AWQ)	2.90 / 0.85	1.60 / 1.64	0.82 / 0.76	3.24 / 2.84	4.56 / 4.24	
Dream-7B-Base (SINQ)	3.67 / 2.56	2.06 / 1.35	1.85 / 1.52	3.39 / 2.29	—	—
Dream-7B-Base (AWQ)	4.10 / 2.13	2.06 / 1.18	1.47 / 0.92	3.08 / 2.05	—	—

Weight Activation Quantization Techniques

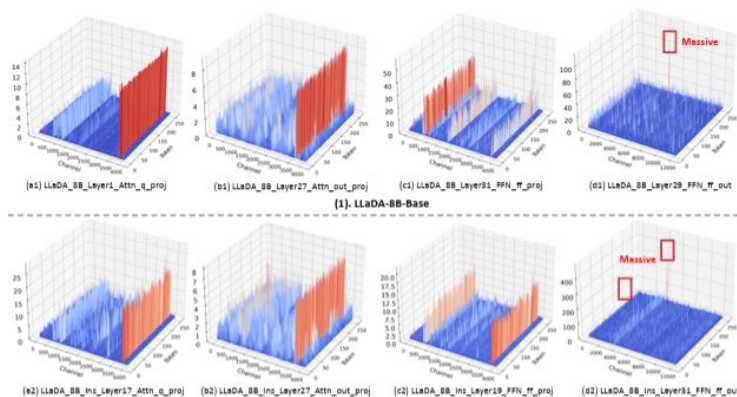
Table 2: Model performance on general tasks under weight-activation quantization.

Model	Setting	Method	WinoGrande	PIQA	ARC-C	ARC-E	MMLU 5-shot	Avg	Drop
LLaDA-8B	FP Model	-	69.9	74.6	46.4	71.1	65.7	65.5	-
	W8A8	SmoothQuant	67.7	70.8	45.5	70.5	65.0	63.9	↓ 2.5%
		QuaRot	68.6	71.1	45.2	70.8	66.1	64.4	↓ 1.8%
		DuQuant	67.9	70.4	45.9	71.4	66.0	64.3	↓ 1.9%
	W4A4	SmoothQuant	49.4	58.8	29.2	40.9	27.1	41.1	↓ 37.3%
		QuaRot	63.4	68.1	43.7	69.2	61.8	59.2	↓ 6.6%
		DuQuant	64.9	69.3	42.8	70.0	64.0	62.2	↓ 5.1%
LLaDA-8B -Instruct	FP Model	-	70.2	71.3	54.3	75.9	64.0	67.1	-
	W8A8	SmoothQuant	69.6	72.1	53.5	75.9	64.0	67.0	↓ 0.2%
		QuaRot	69.1	71.3	54.1	76.2	64.1	67.0	↓ 0.3%
		DuQuant	68.8	71.8	54.6	76.2	63.6	67.0	↓ 0.2%
	W4A4	SmoothQuant	52.3	65.1	34.0	54.0	32.8	47.7	↓ 29.0%
		QuaRot	65.2	69.8	51.3	75.1	61.1	64.5	↓ 3.9%
		DuQuant	66.4	72.2	52.7	74.8	61.2	65.4	↓ 2.5%

H. Lin *et al.*, “Quantization Meets dLLMs: A Systematic Study of Post-training Quantization for Diffusion LLMs,” Oct. 15, 2025, *arXiv*: arXiv:2508.14896. doi: [10.48550/arXiv.2508.14896](https://doi.org/10.48550/arXiv.2508.14896).

DuQuant

- Rotation-based quantization technique
- Uses outlier-aware rotation matrices and channel permutation to better align the activation distribution with quantization-friendly structures.



[1]

H. Lin *et al.*, “DuQuant: Distributing Outliers via Dual Transformation Makes Stronger Quantized LLMs,” Nov. 01, 2024, *arXiv*: arXiv:2406.01721. doi: [10.48550/arXiv.2406.01721](https://doi.org/10.48550/arXiv.2406.01721).

DuQuant + SINQ

DuQuant first redistributes activation outliers before quantization then SINQ applies weight quantization

- How it works
 1. Pass calibration data through the model
 2. Collect activations statistics
 3. Compute rotation matrix using DuQuant
 4. apply rotation matrix to weights
 5. Use SINQ on redistributed weights
 6. At inference time the input x is also rotated ($x @ R$) before the matmul

DuQuant + SINC Results

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The first/second number indicates the % of correct/incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINC)	1.88 / 1.28	1.60 / 1.30	1.03 / 0.49	2.84 / 2.84	3.85 / 4.37	0.64 / 0.28
LLaDA-8B-Base (AWQ)	2.90 / 0.85	1.60 / 1.64	0.82 / 0.76	3.24 / 2.84	4.56 / 4.24	
LLaDA-8B-Base (DuSINC)	1.62/1.19		1.09/0.54	2.84/2.37		
Dream-7B-Base (SINC)	3.67 / 2.56	2.06 / 1.35	1.85 / 1.52	3.39 / 2.29	—	—
Dream-7B-Base (AWQ)	4.10 / 2.13	2.06 / 1.18	1.47 / 0.92	3.08 / 2.05	—	—

Plan for next week

- Report perplexity results with all quantization techniques
- Fix issues with GPTQ
- Implement weight activation quantization (W4A8,W4A4) on duquant and other techniques combined with SINQ and compare the results